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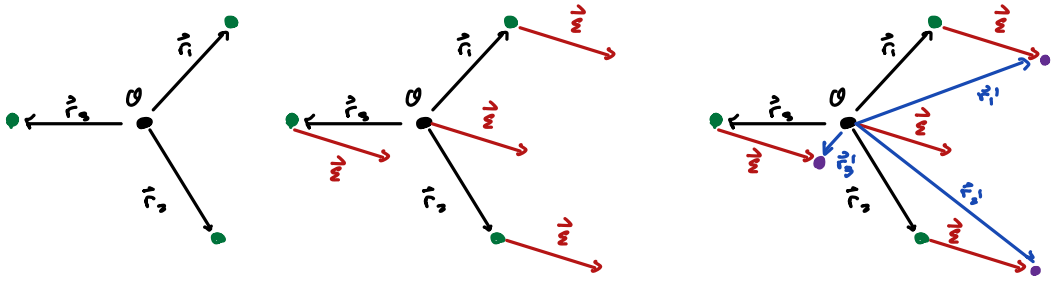
Lecture 34 4/15/26

Conservation Laws & Noether's Theorem

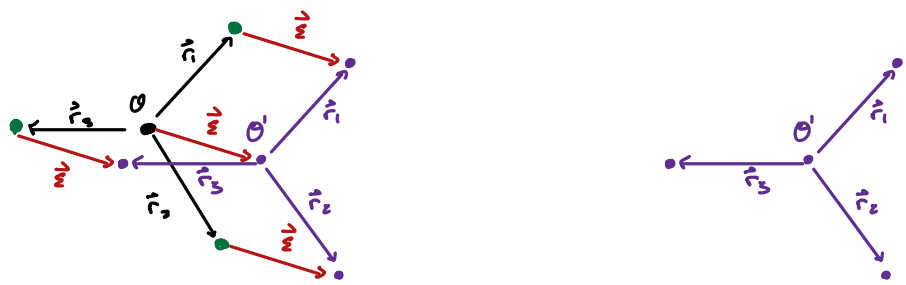
Conservation of momentum $\Leftrightarrow$ translation invariance

$\vec{r}_1 \to \vec{r}_1 + \vec{\varepsilon}$, $\vec{r}_2 \to \vec{r}_2 + \vec{\varepsilon}$, ... , $\vec{r}_N \to \vec{r}_N + \vec{\varepsilon}$



translation is the same as shifting the position of the origin

Consider moving origin:



invariant!

If $u(\vec{r}_1 + \vec{\varepsilon}, \vec{r}_2 + \vec{\varepsilon}, \dots, \vec{r}_N + \vec{\varepsilon}, t) = u(\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N, t)$, then u is invariant under translation

$\Rightarrow \delta u = 0$

We know T is invariant under translation: $\dot{\vec{r}}_\alpha{}' = \frac{d}{dt}(\vec{r}_\alpha{}') = \frac{d}{dt}(\vec{r}_\alpha + \vec{\varepsilon}) = \frac{d}{dt}\vec{r}_\alpha + 0 = \dot{\vec{r}}_\alpha$

So $\delta u = 0 \Rightarrow \delta \mathcal{L} = 0$ under translations

Let $\vec{\varepsilon}$ be an infinitessimal displacement in the $\vec{x}$-direction:

$\mathcal{L} \to \mathcal{L} + \delta\mathcal{L}$ $\qquad x \to x + \varepsilon = x + \delta x$

$$\delta \mathcal{L} = \frac{\partial \mathcal{L}}{\partial x_1}\delta x_1 + \frac{\partial \mathcal{L}}{\partial x_2}\delta x_2 + \dots + \frac{\delta \mathcal{L}}{\partial x_N}\delta x_N$$

$$0 = \varepsilon \frac{\partial \mathcal{L}}{\partial x_1} + \varepsilon \frac{\partial \mathcal{L}}{\partial x_2} + \dots + \varepsilon \frac{\partial \mathcal{L}}{\partial x_N}$$

$$0 = \sum_{\alpha=1}^{N} \frac{\partial \mathcal{L}}{\partial x_\alpha}$$ $\to$ true for each $\frac{\partial \mathcal{L}}{\partial x_\alpha}$ individually

Use Lagrange equation: $\frac{\partial \mathcal{L}}{\partial x_\alpha} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}_\alpha} = \frac{d}{dt} p_{\alpha x}$

$$0 = \sum_{\alpha=1}^{N} \frac{d}{dt} p_{\alpha x} = \frac{d}{dt}\sum_{\alpha=1}^{N} p_{\alpha x} = \boxed{\frac{d}{dt} P_x = 0}$$

Total Momentum is conserved (can show the same in y & z directions)

<u>Conservation of energy $\Longleftrightarrow$ time translation invariance</u>

In general, $\mathcal{L}(q_1, q_2, \dots, \dot{q}_1, \dot{q}_2, \dots, t)$ changes with time as t, q_i, $\dot{q}_i$ change

$$\frac{d\mathcal{L}}{dt} = \frac{\partial \mathcal{L}}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\frac{d\dot{q}_i}{dt} + \frac{\partial \mathcal{L}}{\partial t}\frac{dt}{dt} \qquad \text{(sum over } i\text{)}$$

$$= \frac{\partial \mathcal{L}}{\partial q_i}\dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i}\ddot{q}_i + \frac{\partial \mathcal{L}}{\partial t}$$

We know $\frac{\partial \mathcal{L}}{\partial q_i} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{d}{dt} p_i = \dot{p}_i$

generalized momentum

$$\frac{d\mathcal{L}}{dt} = \underbrace{\dot{p}_i q_i + p_i \ddot{q}_i}_{\text{product rule}} + \frac{\partial \mathcal{L}}{\partial t}$$

$$\frac{d\mathcal{L}}{dt} = \frac{d}{dt}\left(p_i \dot{q}_i\right) + \frac{\partial \mathcal{L}}{\partial t}$$

$$0 = \frac{d}{dt}\left(p_i \dot{q}_i - \mathcal{L}\right) + \frac{\partial \mathcal{L}}{\partial t}$$

If $\frac{\partial \mathcal{L}}{\partial t} = 0 \Rightarrow \frac{d}{dt}\left(p_i \dot{q}_i - \mathcal{L}\right) = 0$

Define $\mathcal{H} \equiv p_i \dot{q}_i - \mathcal{L} \equiv$ Hamiltonian

<u>If $\mathcal{L}$ does not depend explicitly on time $\left(\frac{\partial \mathcal{L}}{\partial t} = 0\right)$, then $\mathcal{H}$ is conserved</u> !

This conservation law is important: if the generalized coordinates are <u>natural</u>:

$\vec{r}_\alpha = \vec{r}_\alpha(q_1, \dots, q_n)$ (no explicit time-dependence)

then $\mathcal{H} = T + U =$ total energy.

Let's show this:

$$\vec{r}_\alpha = \vec{r}_\alpha(q_1, \dots, q_n)$$

$$\dot{\vec{r}}_\alpha = \frac{d}{dt}(\vec{r}_\alpha) = \frac{\partial \vec{r}_\alpha}{\partial t}\frac{d}{dt}q_1 + \frac{\partial \vec{r}_\alpha}{\partial q_2}\frac{dq_2}{dt} + \dots + \frac{\partial \vec{r}_\alpha}{\partial q_n}\frac{dq_n}{dt} = \frac{\partial \vec{r}_\alpha}{\partial q_i}\frac{dq_i}{dt} = \frac{\partial \vec{r}_\alpha}{\partial q_i}\dot{q}_i$$

$$\dot{\vec{r}}_\alpha^{\,2} = \dot{\vec{r}}_\alpha \cdot \dot{\vec{r}}_\alpha = \left[\frac{\partial \vec{r}_\alpha}{\partial q_i}\dot{q}_i\right]\cdot\left[\frac{\partial \vec{r}_\alpha}{\partial q_j}\dot{q}_j\right] = \frac{\partial \vec{r}_\alpha}{\partial q_i}\cdot\frac{\partial \vec{r}_\alpha}{\partial q_j}\dot{q}_i\dot{q}_j$$

$$T = \frac{1}{2}m_\alpha \dot{\vec{r}}_\alpha^{\,2} = \frac{1}{2}\sum_\alpha m_\alpha\left(\frac{\partial \vec{r}_\alpha}{\partial q_i}\right)\cdot\left(\frac{\partial \vec{r}_\alpha}{\partial q_j}\right)\dot{q}_i\dot{q}_j$$

$$p_i = \frac{\partial \mathcal{L}}{\partial \dot{q}_i} = \frac{\partial T}{\partial \dot{q}_i} = \sum_\alpha m_\alpha\left(\frac{\partial \vec{r}_\alpha}{\partial q_i}\right)\cdot\left(\frac{\partial \vec{r}_\alpha}{\partial q_j}\right)\dot{q}_j$$

We want $p_i\dot{q}_i$:

$$p_i\dot{q}_i = \sum_\alpha m_\alpha\left(\frac{\partial \vec{r}_\alpha}{\partial q_i}\right)\cdot\left(\frac{\partial \vec{r}_\alpha}{\partial q_j}\right)\dot{q}_j\dot{q}_i = 2T$$

$$\mathcal{H} = p_i\dot{q}_i - \mathcal{L} = 2T - (T-U) = 2T - T + U = T + U = E_T\,!$$

$\Rightarrow$ If the generalized coordinates are natural and $\frac{\partial \mathcal{L}}{\partial t} = 0$,

then E_T is conserved!

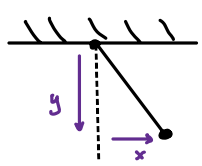
Lagrange Multipliers & Constraint forces

When we use the minimal # of generalized coordinates, we sidestep finding constraint forces (like F_T, F_N, F_{fs}, etc.), but sometimes we want to find them!

$\Rightarrow$ Use the method of Lagrange multipliers.

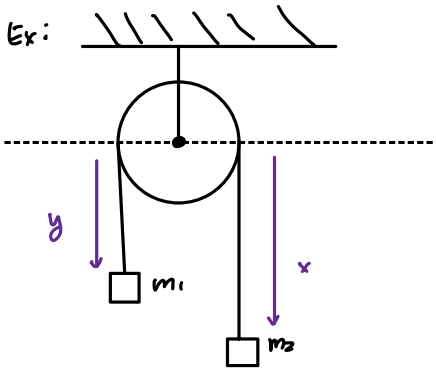
Consider a constrained system $f(x,y) = \text{constant}$

Ex:



$$\sqrt{x^2+y^2} = \ell \Rightarrow f(x,y) = \sqrt{x^2+y^2}$$

Ex:

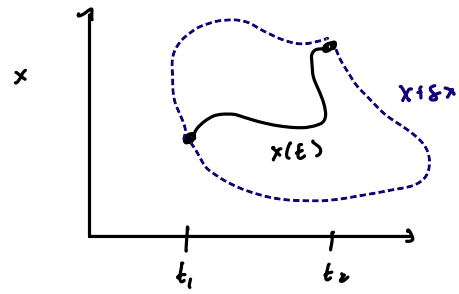


$$x + y + \pi R = l$$
$$x + y = l - \pi R \quad \Rightarrow \quad f(x,y) = x + y$$

Back to Hamilton's Principle:

$$S = \int_{t_1}^{t_2} \mathcal{L}(x, \dot{x}, y, \dot{y})\, dt$$

$$x(t) \to x(t) + \delta x(t)$$
$$\underbrace{y(t)}_{\text{stationary path}} \to \underbrace{y(t) + \delta y(t)}_{\text{wrong path}}$$



$$\delta S = \int \left(\frac{\partial \mathcal{L}}{\partial x} \delta x + \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x} + \frac{\partial \mathcal{L}}{\partial y} \delta y + \frac{\partial \mathcal{L}}{\partial \dot{y}} \delta \dot{y} \right) dt = 0 \qquad \text{(chain rule)}$$

integrate by parts

$$\int \frac{\partial \mathcal{L}}{\partial \dot{x}} \delta \dot{x}\, dt = \int \frac{\partial \mathcal{L}}{\partial \dot{x}} \frac{d}{dt}(\delta x)\, dt = \cancel{\text{boundary term}}^{\,0} - \int \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{x}}\right) \delta x\, dt$$

$\delta x = 0$ at t_1, t_2

$$\delta S = \int \left(\frac{\partial \mathcal{L}}{\partial x} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{x}} \right) \delta x\, dt + \int \left(\frac{\partial \mathcal{L}}{\partial y} - \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{y}} \right) \delta y\, dt = 0$$

[Requirement for S = stationary: $\delta S = 0$]

Before, we said δx = arbitrary, not zero
δy = arbitrary, not zero $\Rightarrow$ two independent Lagrange equations

Not true anymore!

$f(x,y)$ = constant <u>constrains</u> $\delta x, \delta y$

$$\delta f = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y = 0$$

Since $\delta f = 0$, then $\lambda(t)\delta f$ is zero for any function $\lambda(t)$

So $\delta S = 0 \Rightarrow \delta S + \lambda \delta f = 0$ [just $\delta S + 0 = 0$]

$$\delta f = \frac{\partial f}{\partial x}\delta x + \frac{\partial f}{\partial y}\delta y = 0$$

$$\delta S + \lambda \delta f = \int \underbrace{\left(\frac{\partial \mathcal{L}}{\partial x} + \lambda \frac{\partial f}{\partial x} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}\right)}_{I_x} \delta x \, dt + \int \underbrace{\left(\frac{\partial \mathcal{L}}{\partial y} + \frac{\lambda \partial f}{\partial y} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}\right)}_{I_y} \delta y \, dt = 0$$

We are free to choose $\lambda(t)$ to force $I_x = 0$ $\Rightarrow$ $\boxed{\frac{\partial \mathcal{L}}{\partial x} + \lambda \frac{\partial f}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}}}$ defines "correct" $\lambda(t)$

Here's the trick:

$$\delta S = 0 = \delta S + \lambda \delta f = \int I_1 \, dx + \int I_2 \, dx = 0$$

We pick $\lambda(t)$ s.t. $I_1 = 0$

$$\Rightarrow \int I_2 \, dt = 0$$

$$\int \left(\frac{\partial \mathcal{L}}{\partial y} + \lambda \frac{\partial f}{\partial y} - \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}\right)\delta y \, dt = 0$$

This can now vary independently. It's just tied to δx, but δx is no longer in the equation

$$\Rightarrow \left.\begin{array}{l} \boxed{\frac{\partial \mathcal{L}}{\partial y} + \lambda \frac{\partial f}{\partial y} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}} \\ \boxed{\frac{\partial \mathcal{L}}{\partial x} + \lambda \frac{\partial f}{\partial x} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}} \\ \boxed{f(x,y) = \text{const}} \end{array}\right\}$$

These 3 equations give EOM's & $\lambda(t)$

$$\mathcal{L} = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}m\dot{y}^2 - U(x,y)$$

$$-\frac{\partial U}{\partial x} + \lambda \frac{\partial f}{\partial x} = m_1 \ddot{x}$$

$$F_x + \lambda \frac{\partial f}{\partial x} = m_1 a = \sum F_{total} = F_x + F_x^{constraint}$$

$$\Rightarrow \lambda \frac{\partial f}{\partial x} = F_x^{constraint}$$

$\Rightarrow$ you can find the constraint force!